BIOINFORMATICS DATA ENGINEERING / PHD RESEARCH SYNOPSIS

CancerOmicsLake

Reproducibility and the sensitivity of candidate-gene rankings

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Project repository and source code

1. Question and contribution

Does agreement across a broad gene-expression profile justify confidence in its leading candidate genes? We built CancerOmicsLake to measure how reference choice, processing, ties and numerical execution affect that inference. The main empirical result is that moderate signed-effect concordance coexists with very weak top-50 overlap in exactly matched native/recount3 comparisons. The contribution combines an implemented data platform with an auditable sensitivity evaluation; processing variability itself is established prior work [1, 2].

System and data

The platform integrates open-access GDC/TCGA expression, mutation and metadata with GTEx normals, gene annotations and public recount3 processed expression. It covers breast invasive carcinoma (BRCA), lung adenocarcinoma (LUAD) and colon adenocarcinoma (COAD), with breast, lung and two colon reference tissues. The frozen expression inventory contains **36,456,660 TCGA** and **11,240,000 GTEx** measurement rows. These are gene–specimen values, not counts of patients or sequencing files.

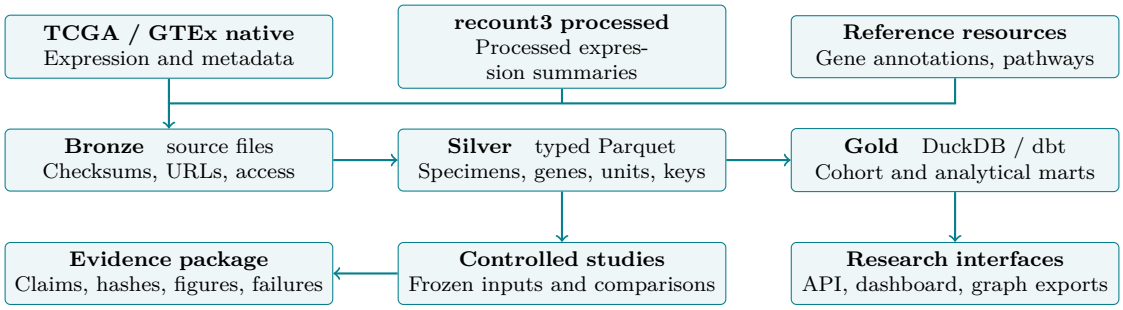


Figure 1. Implemented workflow. Acquisition preserves source context; harmonization validates entities and measurements before analytical use. Frozen study inputs connect analytical outputs to claim-level evidence. Arrows denote data dependencies, not validation of biological truth.

What was implemented and evaluated

Python/Polars transformations, DuckDB/dbt models, orchestration, Docker and CI support configured ingestion, bounded downloads, retries and validation. Streamlit, FastAPI and Neo4j/Graphify exports expose the analytical layers. Exploratory analytics include consequence-stratified mutation frequencies, expression comparisons, candidate rankings, pathway overrepresentation with within-cancer/candidate-set BH correction, and graph degree summaries.

The evaluated contribution has three parts: (i) specimen/gene harmonization and source-to-claim traceability; (ii) matched processing and reference-sensitivity studies with uncertainty diagnostics; and (iii) numerical replay and external baseline/failure tests. Repairs addressed duplicate keys, preservation of `_PAR_Y` loci and donor/case leakage. Earlier evidence is retained when later designs change. Mutation, pathway and graph summaries provide context; they are not independent biological validation of the rankings.

Interpretation. The platform makes the conditions behind a candidate list inspectable. The research evaluates its reliability and limitations rather than treating a successful execution as evidence of a stable scientific conclusion.

2. Matched processing study: design and principal result

Exact identity, case/donor, tissue, QC and count-column checks yielded **617 available specimens**: 417 tumors and 200 GTEx normals. The three primary contrasts use **567 specimens** (417 tumors plus 150 normals); 50 sigmoid-colon normals are retained separately. COAD's primary reference is transverse colon. V2 followed inspection of v1 and is an exploratory diagnostic study.

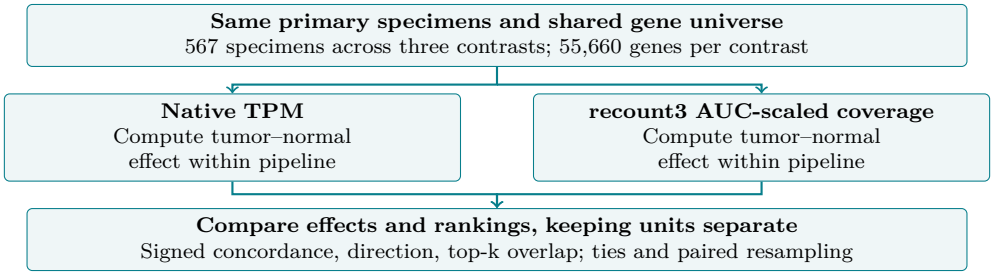


Figure 2. Matched comparison. Membership is fixed across pipelines. Each gene's rank-biserial effect is the empirical probability that a tumor value exceeds a normal value minus the reverse probability, ranging from −1 to 1. This rank-based contrast does not make numerical units equivalent.

Cancer	Tumors	GTEx normals	Direction agreement	Eligible genes
BRCA	162	50	89.85%	23,247
COAD	119	50	90.53%	22,007
LUAD	136	50	90.21%	23,732

Direction agreement is conditional on $|r_g| \geq 0.2$ in both pipelines. Its gene denominator is shown; it is not a proportion of specimens.

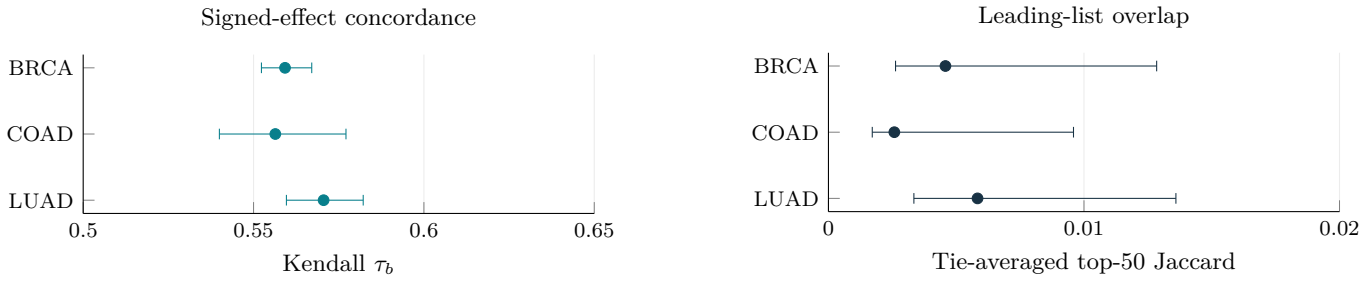


Figure 3. Broad agreement coexists with weak leading lists. Points show observed τ_b or mean tie-resolved Jaccard; bars show descriptive 95% paired case/donor bootstrap percentile intervals (200 replicates). Top-50 lists rank absolute effects. Axes use different ranges and statistics; magnitudes are not directly comparable.

Across all shared genes, τ_b was 0.556–0.571 while mean top-50 Jaccard was 0.0026–0.0058. Native cutoff ties spanned 412–515 genes. V2 separately reported tie-only variability (1,000 seeded resolutions) and bootstrap intervals with 100 tie resolutions per replicate. A fixed 20% detection filter in both groups/pipelines improved broad concordance but did not rescue short lists. These descriptive results are conditional on available matched specimens; processing and annotation causes were not experimentally separated.

3. Reference and reproducibility studies

Does a more elaborate score beat a simple reference baseline?

Separate corrected native and paired reference studies compared adjacent-normal and GTEx controls with globally separated case/donor partitions. Within each partition, tumor specimens and reference sizes were matched across methods. Selected development/holdout counts *per group* were BRCA 26/24, COAD 21/20, LUAD 26/24 in each design. These smaller ranking cohorts differ from the processing study on p. 2; paired designs additionally constrain tumor/adjacent-normal case matching.

A bootstrap lower-quantile score was compared with single-reference effects, mean ranks, maximin and the official R `RobustRankAggreg` implementation [3]. Top-25/50/100 evaluation used heldout effect/direction summaries and paired resampling; reference ablations and bootstrap rank stability were exploratory diagnostics. The proposed score had **no top-50 advantage over maximin**. RRA was evaluated as a ranking: shared tumor specimens invalidate an independent-list significance interpretation. Constructed shared-normal bias also produced reference agreement with incorrect biological selections.

What reproduced, and what failed the original numerical gate?

Archived replay compared macOS/Linux with pinned dependencies and Polars 1.42.1/1.44.2, retaining rtol = 10^{-8} and atol = 10^{-10}. Ordering, ties and intermediate arrays were recorded. Three historical failed rows repeated one Spearman statistic at three top-k values. The earliest observed divergence was in log2 effect arrays; the underlying platform cause was not established by varying dependencies.

Candidate lists and heldout metrics reproduced in the corrected global studies. However, **63/8,100 native and 60/8,100 paired values** of the `abs_effect_spearman` sensitivity column exceeded tolerance. These are failed scalar comparisons, not failed specimens or entire candidate lists. Existing eager rational, specialized rational and CGAL lazy exact methods later agreed in all 46 workload/specification comparisons, including native macOS checks. Of 6,624 process attempts, 36 binary64 attempts failed on a constructed extreme-value fixture. This later study did not repair the historical floating-point acceptance failures.

External preparation comparison and failure recovery

The pinned `tcga-data-nf` implementation [4] was compared on cached recount3 data with matching membership and normalization/filter semantics. Outputs matched in 21 preparation workloads and 7 native-script integrations. Equivalence preceded timing. The shared R normalization backend, added adapter and emulated hardware restrict performance interpretation; end-to-end Nextflow and intrinsic engine-speed superiority were not established.

Test	Observed result	What it establishes
Corrupt input, missing metadata, conflicting duplicates	Benchmark adapter/parser rejected each	Detection at the recorded validation layer.
Interrupted execution	No incorrect output committed; fresh retry recovered	Recovery with an explicit pause hook, not native automatic resume.
Stale cached evidence	Evidence hash guard rejected reuse	Local evidence integrity; external native handling unmeasured.
Frozen sources and reports	Inputs, plans and earlier failures retained	An audit trail, not authentication of original biological assertions.

Interpretation. The evidence supports controlled comparison and traceability. It also records numerical failures and the absence of ranking/performance superiority; passing software tests does not cancel these scientific limitations.

4. Method experiments and research scope

Additional exploratory studies tested whether the observed failure modes motivated a distinct method or systems benefit. Each addressed a bounded question and used declared comparators. None established a superior new method.

Study	Executed evidence	Result and interpretation
Evidence contracts	32 synthetic cases; 192 tool/case/pass decisions	SQL, SHACL and workflow gating matched the prototype. All admitted two false attestations.
Decision-aware recomputation	400 distinct patch events in 40 workloads; 233 unchanged sets certified	No decision mismatches; lifecycle cost beat exact incremental updating in 0/40 workloads.
Exact ranking arithmetic	46 comparisons of established exact methods	Agreement demonstrated; no unique capability or uniformly fastest implementation.
Selection with abstention	3,600 synthetic datasets; Holm, Bonferroni and joint bootstrap	The narrow top-k label gate passed, but invalid pair directions occurred in 47/200 versus 5/200 datasets in one moderate-signal cell.
Minimum claim-flipping changes	112 synthetic alternatives in 7 scenarios	SQL and Python agreed on all decisions and minimum-change witnesses. Actual quantification-workflow generation was untested.

Repeated timings/tool passes are not independent datasets. The abstention example uses 30 donors. Its pairwise error prevents a calibrated confidence claim despite correct observed top-k labels. The official confident feature-ranking implementation [5] was not benchmarked.

Relationship to the literature

Arora et al. [1] already established that pipeline choice affects expression and downstream results. Recount3 [2] supplies uniformly processed expression; RRA [3] and confident feature ranks [5] provide established ranking methods. CancerOmicsLake's contribution is an implemented, auditable case study linking matched effect rankings, tie uncertainty, numerical replay and failure-aware evidence packaging.

Arora replication remains unexecuted: checked expression-archive locations returned access errors on 16 October 2026. This does not show that the matrices were never published. Their endpoints and ours are not a matched head-to-head benchmark. The external preparation comparison has the scope stated on p. 3.

Scope of inference and remaining work

The project demonstrates biomedical data engineering and a substantive sensitivity evaluation. Its strongest finding is the distinction between broad effect agreement, short-list stability and numerical acceptance. Annotation, processing, collection and batch effects were not causally separated. Only three cancers and available matched specimens were examined; inspected holdouts are development evidence. Neither clinical utility nor biological validation has been demonstrated.

The next requirements are independent investigator reproduction of the prepared handoff, external evaluation on identical endpoints and fresh evidence selected before further method tuning. A future algorithm would need an explicit estimand and improvement over competent baselines, including maximin. This platform is an implemented starting point for my PhD interest in reliable biomedical data infrastructure and reproducible analysis.

Status. Four-page research synopsis for interview discussion; detailed methods and evidence accompany the repository. AI tools assisted implementation and drafting. Author review and formal declarations remain to be finalized; independent human reproduction is pending.

References

[1] Arora et al. *Variability in estimated gene expression among commonly used RNA-seq pipelines*. Scientific Reports, 2020. doi:10.1038/s41598-020-59516-z.
[2] Wilks et al. *recount3: summaries and queries for large-scale RNA-seq expression and splicing*. Genome Biology, 2021. doi:10.1186/s13059-021-02533-6.
[3] Kolde et al. *Robust rank aggregation for gene list integration and meta-analysis*. Bioinformatics, 2012. doi:10.1093/bioinformatics/btr709.
[4] Quackenbush Lab. *tcga-data-nf*. Pinned workflow source; limited comparison on p. 3.
[5] Neuhof and Benjamini. *Confident Feature Ranking*. AISTATS, 2024. PMLR 238:1468–1476.